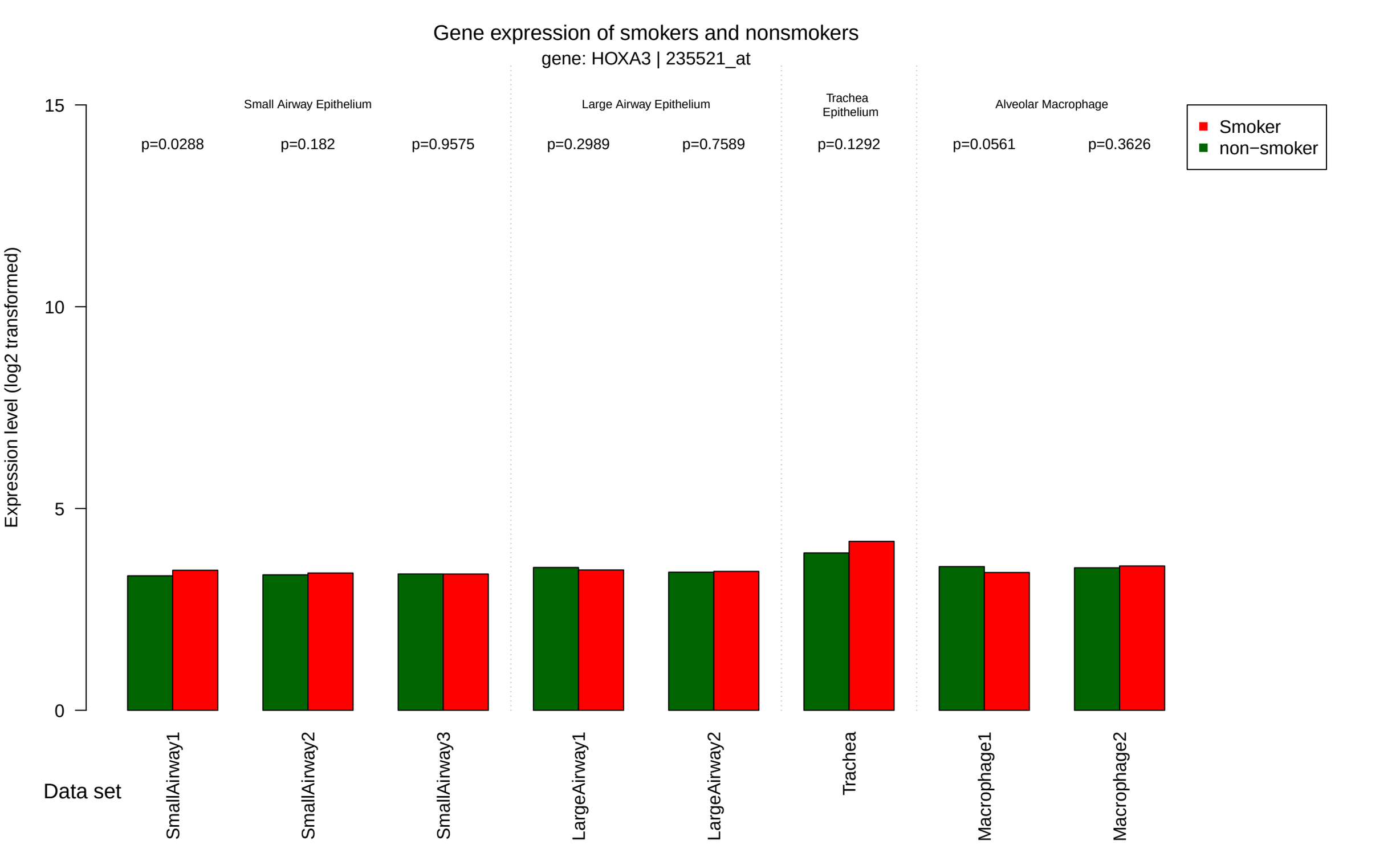
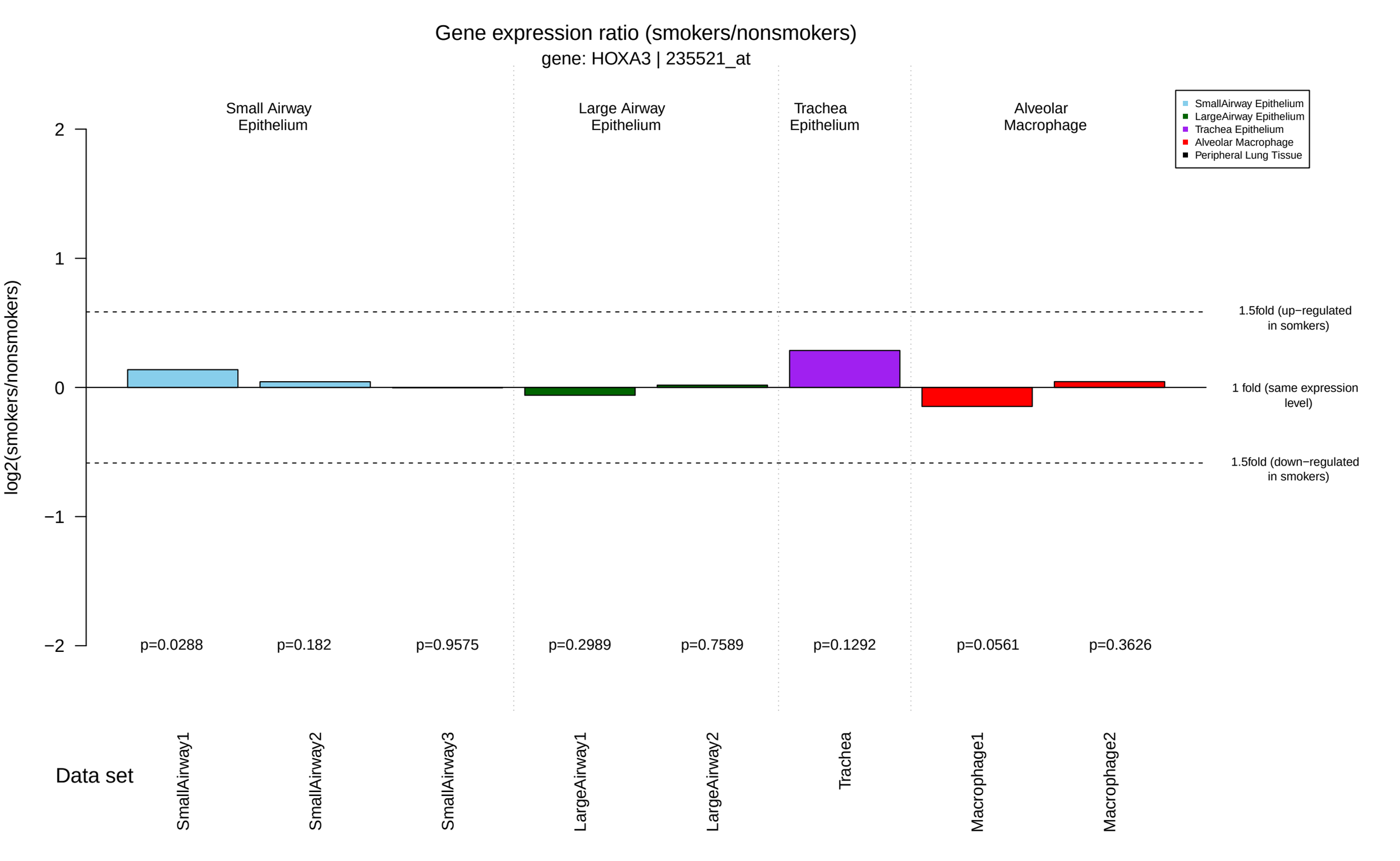
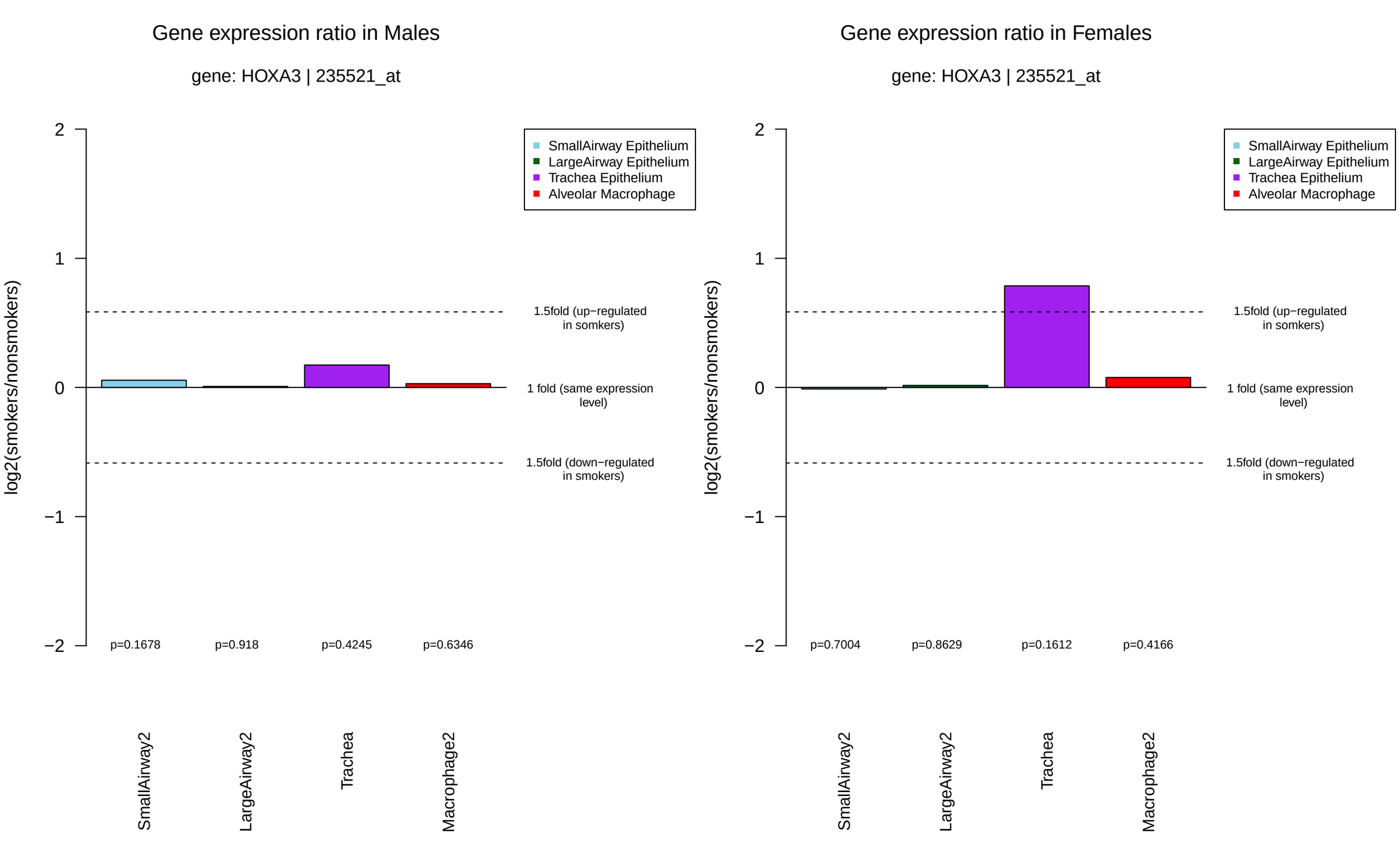
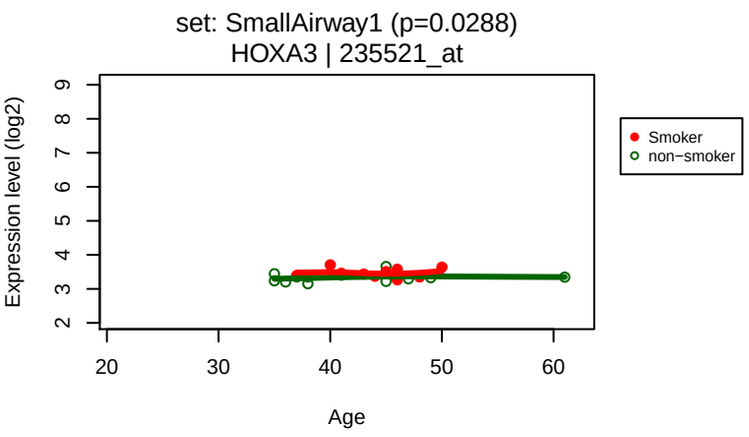
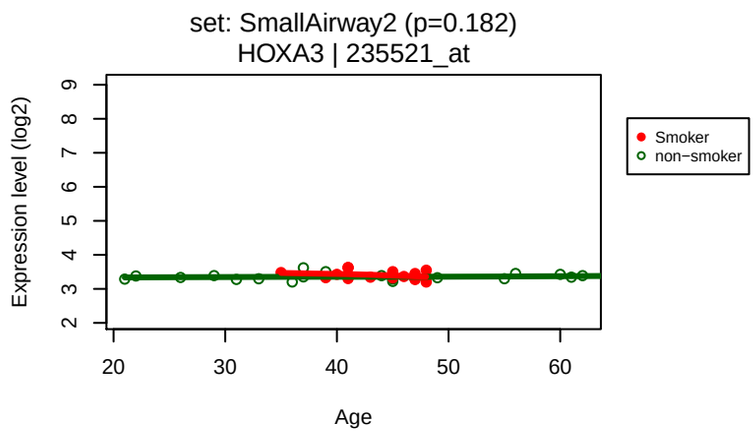
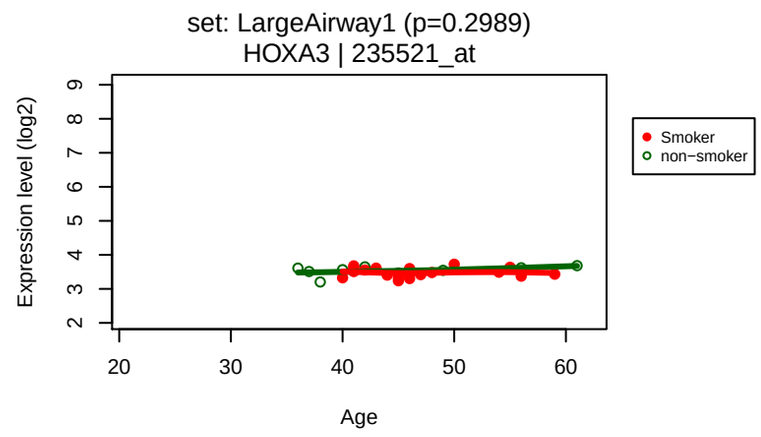

Gene expression of smokers and nonsmokers
gene: HOXA3 | 235521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0288
p=0.182
p=0.9575
p=0.2989
p=0.7589
p=0.1292
p=0.0561
p=0.3626
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

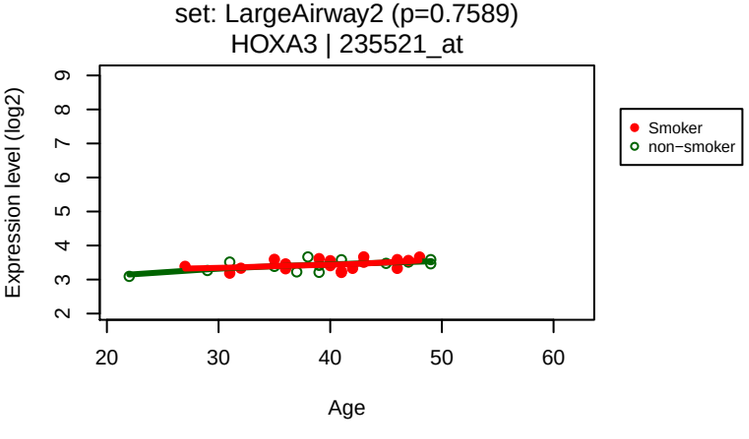

Gene expression ratio (smokers/nonsmokers)
gene: HOXA3 | 235521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0288
p=0.182
p=0.9575
p=0.2989
p=0.7589
p=0.1292
p=0.0561
p=0.3626
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

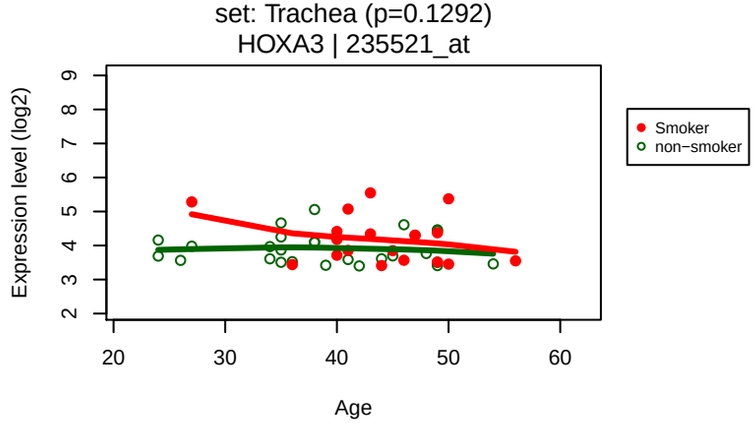

Gene expression ratio in Males
gene: HOXA3 | 235521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1678
p=0.918
p=0.4245
p=0.6346
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA3 | 235521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7004
p=0.8629
p=0.1612
p=0.4166
SmallAirway2
LargeAirway2
Trachea
Macrophage2

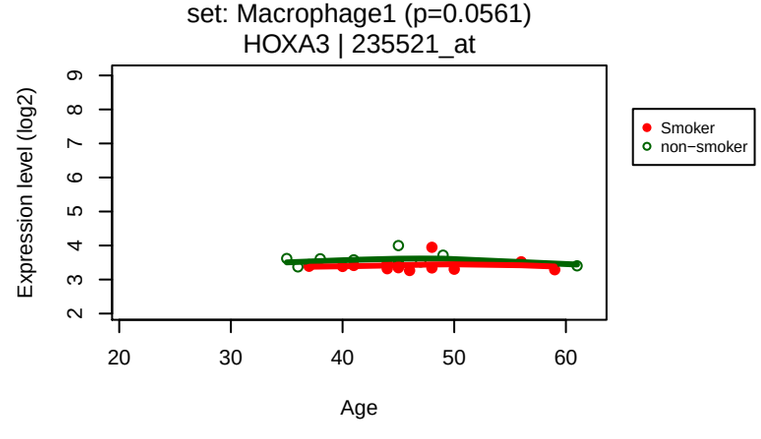

set: SmallAirway1 (p=0.0288)
HOXA3 | 235521_at
Expression level (log2)
Age
Smoker
non−smoker

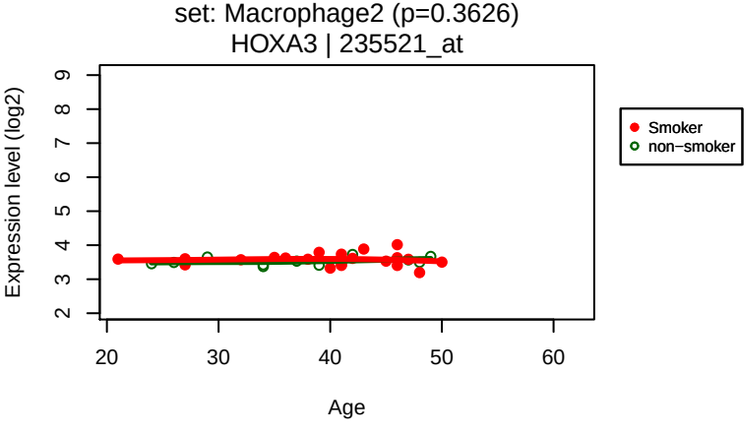

set: SmallAirway2 (p=0.182)
HOXA3 | 235521_at
Expression level (log2)
Age
Smoker
non−smoker

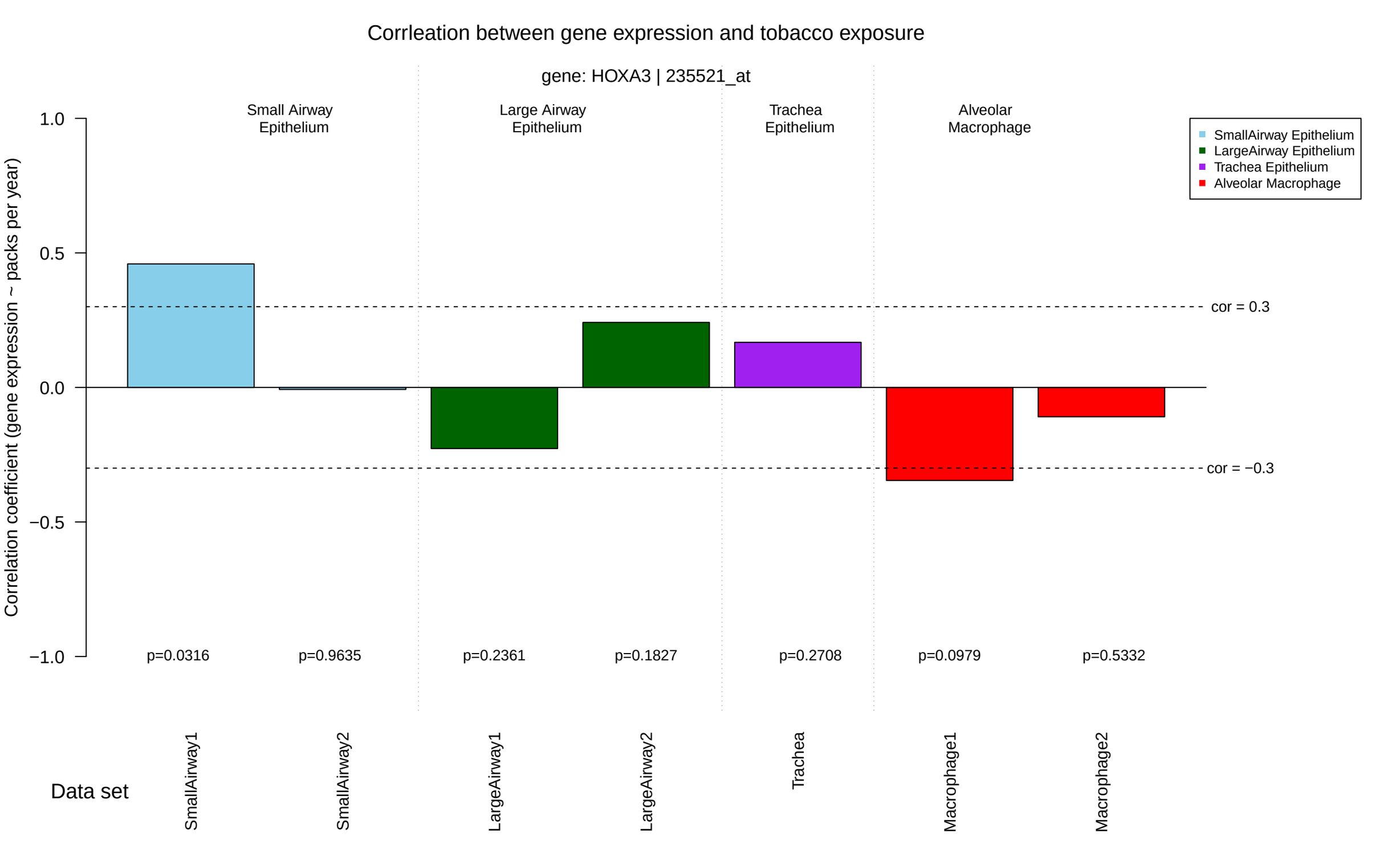

set: LargeAirway1 (p=0.2989)
HOXA3 | 235521_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7589)
HOXA3 | 235521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1292)
HOXA3 | 235521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0561)
HOXA3 | 235521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3626)
HOXA3 | 235521_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA3 | 235521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0316
p=0.9635
p=0.2361
p=0.1827
p=0.2708
p=0.0979
p=0.5332
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2